Colorimeter Prototype
Pin Mapping and Configuration Reference

Raspberry Pi Zero 2 W — As-Built

Pre-testing reference document. Describes all hardware connections, required OS configuration, known issues, and testing cautions as of first prototype build.

RPi GPIO reference: raspberrypi.com/documentation/.../raspberry-pi.html#gpio
Adafruit 1651: learn.adafruit.com/adafruit-2-8-tft-touch-shield-v2
74HC595: TI SN74HC595 datasheet; learn.adafruit.com/74hc595
pigpio library: abyz.me.uk/rpi/pigpio

RPi Zero 2 W GPIO Peripheral Reference

All GPIO pins operate at **3.3 V logic**. Applying 5 V to any GPIO pin will permanently damage the SoC. The following peripheral pin assignments are **fixed by the RP3A0 SoC** and cannot be remapped in software.

Source: raspberrypi.com — GPIO and 40-pin header documentation

Hardware PWM — available on four pins only: GPIO12 (PWM0-A, pin 32), GPIO13 (PWM1, pin 33), GPIO18 (PWM0-B, pin 12), GPIO19 (PWM0, pin 35). *GPIO12 and GPIO13 share one clock divider. GPIO18 and GPIO19 share another. Both channels within a pair can have independent duty cycles but their frequencies must be derived from the same base clock.*

SPI0 (enabled by `dtparam=spi=on`): MOSI=GPIO10/pin 19; MISO=GPIO9/pin 21; SCLK=GPIO11/pin 23; CE0=GPIO8/pin 24; CE1=GPIO7/pin 26.

SPI1 (enabled by `dtoverlay=spi1-1cs`): MOSI=GPIO20/pin 38; MISO=GPIO19/pin 35; SCLK=GPIO21/pin 40 CE0=GPIO18/pin 12; CE1=GPIO17/pin 11; CE2=GPIO16/pin 36. *Warning: the spi1 overlay claims GPIO18 as CE0 by default. GPIO18 is used in this design for a separate purpose. See Section 5.2 for correct configuration.*

I2C bus 1 (enabled by `dtparam=i2c_arm=on`): SDA=GPIO2/pin 3; SCL=GPIO3/pin 5.

UART: TX=GPIO14/pin 8; RX=GPIO15/pin 10. *Reserved for Linux serial console. Do not use.*

EEPROM: GPIO0/pin 27 and GPIO1/pin 28. *Physical pins removed on this prototype. Do not use.*

Power and Ground Pin Notes

5 V pins — Pin 2 and Pin 4 (even side): Electrically identical. Both connect directly to the micro-USB supply rail. In this design: Pin 2 supplies the Adafruit 1651 shield via J1. Pin 4 supplies the LED load circuit via J5–J8. Current limit is set by the micro-USB supply and the board polyfuse (≈2 A for the Zero 2 W).

3.3 V pins — Pin 1 (odd) and Pin 17 (odd): Both are outputs of the same onboard 3.3 V regulator. Electrically identical. **Total current budget across the entire 3.3 V rail is ≈50 mA**, shared among GPIO output drivers and all 3.3 V peripherals. In this design the 3.3 V rail supplies: 74HC595 VCC, 1-Wire pull-up, I2C pull-ups (via 1651 shield), and J2 pin 1 (1-Wire device supply). The 1651 shield has its own onboard 3.3 V regulator fed from 5 V — it does *not* draw from the RPi 3.3 V rail.

Ground: Pins 6, 9, 14, 20, 25, 30, 34, 39 are all identical ground. Use the ground pin nearest to each signal group for lowest-impedance return paths.

Complete RPi Zero 2 W Pin Allocation — As-Built

Legend: Odd-side pin (1651 / SPI0 / I2C) Even-side pin (general I/O) Power / Ground Note / caution

Columns: Phys = physical header pin; BCM = Broadcom GPIO number (use in Python); Dir = O output / I input / IO bidirectional / PWM / PWR; Connector reference in `[brackets]`.

Phys	BCM	Dir	Assignment	Notes / connector
Odd-side pins — Adafruit 1651 display/touch shield (SPI0, I2C, PWM backlight)				
1	—	PWI	3.3 V	Not connected to 1651 (shield has own regulator). Available for low-current 3.3 V loads (≤50 mA total rail).
3	GPIO2	IO	I2C SDA → TSC2007	**FIXED** by SoC. I2C bus 1 data. Shield has onboard 4.7 kΩ pull-up — do not add PCB pull-up. 33 Ω series ESD resistor on PCB. `[J1-Pin2]`
5	GPIO3	IO	I2C SCL → TSC2007	**FIXED** by SoC. I2C bus 1 clock. Same pull-up note as SDA. 33 Ω series ESD resistor on PCB. `[J1-Pin4]`
7	GPIO4	I	TSIRQ — touch interrupt	Active-low interrupt output from TSC2007. Enable RPi internal pull-up in software: `GPIO.setup(4, GPIO.IN, pull_up_down=GPIO.PUD_UP)`. 33 Ω series ESD resistor on PCB. `[J1-Pin10]`
9	—	GNI	Ground	Odd-side ground return. Connects to J1-Pin3, J1-Pin5, J1-Pin15. `[J1-Pin3, J1-Pin5, J1-Pin15]`
11	GPIO17	—	Spare	Unassigned. Routed to J1 for future use. 33 Ω series resistor on PCB. `[J1-Pin14]`
13	GPIO27	—	No connect	Unassigned. Physical pin present but no PCB connection.
15	GPIO22	—	No connect	Unassigned. Physical pin present but no PCB connection.
17	—	PWI	3.3 V	Not connected to 1651. Same rail as pin 1. Available for low-current 3.3 V loads.
19	GPIO10	O	SPI0 MOSI → ILI9341	**FIXED** by SoC (SPI0 MOSI). High-speed serial data to display controller. 33 Ω series resistor on PCB. `[J1-Pin13]`
21	GPIO9	I	SPI0 MISO ← ILI9341	**FIXED** by SoC (SPI0 MISO). Read-back from display and optional SD card. 33 Ω series resistor on PCB. `[J1-Pin12]`
23	GPIO11	O	SPI0 SCLK → ILI9341	**FIXED** by SoC (SPI0 clock). 33 Ω series resistor on PCB. `[J1-Pin11]`
25	—	GNI	Ground	Additional odd-side ground. `[J1-Pin15 shared]`
27	GPIO0	—	PIN REMOVED — DO NOT USE	HAT ID EEPROM pin. Physical pin removed from header on this prototype.

Phys	BCM	Dir	Assignment	Notes / connector
29	GPIO5	—	Spare / SD Card CS	Unassigned. May be used as active-low chip select for microSD slot on 1651 shield if SD card use is implemented. 33 Ω series resistor on PCB. `[J1-Pin8]`
31	GPIO6	O	TFT DC (Data/Command)	Data/Command select line for ILI9341. High = data byte; Low = command byte. Driven by software before each SPI0 transaction. 33 Ω series resistor on PCB. `[J1-Pin7]`
33	GPIO13	PWM	Backlight PWM → 1651 LITE	**Hardware PWM channel 1 (PWM1).** Controls display backlight brightness via 1651 LITE pin. Solder the LITE/#3 jumper on the 1651 shield to enable. 33 Ω series resistor on PCB. See Section 5.5 for shared clock constraints with GPIO12. `[J1-Pin9]`
35	GPIO19	I	SPI1 MISO (breakout)	SPI1 MISO pin. Not currently connected to any active device. Exposed at J4 for future use (e.g. a second SPI1 peripheral with MISO). Note: GPIO19 is also hardware PWM capable — not used for PWM here. `[J4-Pin4]`
37	GPIO26	—	Spare	Unassigned. Routed to J1 for future use. `[J1-Pin16]`
39	—	GND	Ground	Odd-side ground.
Even-side pins — general I/O: MOSFETs, SPI1 shift register, buzzer, 1-Wire, interface headers				
2	—	PWR	5 V → 1651 VIN	Powers Adafruit 1651 shield via J1. Shield's onboard regulator converts to 3.3 V for ILI9341 and TSC2007. **Do not use RPi 3.3 V rail for 1651 power.** 100 nF ceramic + 10 µF tantalum decoupling at J1 (fitted). `[J1-Pin1]`
4	—	PWR	5 V → LED loads	Supplies 5 V to LED/MOSFET drain circuits via J5–J8 Pin 1. 100 nF ceramic decoupling near each LED bank (fitted). `[J5-Pin1, J6-Pin1, J7-Pin1, J8-Pin1]`
6	—	GND	Ground	Even-side primary ground return.
8	GPIO14	—	UART TX — DO NOT USE	Linux serial console TX. Leave unconnected.
10	GPIO15	—	UART RX — DO NOT USE	Linux serial console RX. Leave unconnected.
12	GPIO18	—	Dual-purpose via jumpers	GPIO18 routes to two destinations via solder jumpers. **JP2 (closed):** GPIO18 → J3-Pin4 (interface header). **JP1 (open):** GPIO18 → J4-Pin6 (interface header). Current state: JP2 closed, JP1 open. GPIO18 is also SPI1 CE0 — see Section 5.2. **Warning:** closing both JP1 and JP2 simultaneously shorts J3-Pin4 to J4-Pin6. Do not close both jumpers. `[JP1, JP2, J3-Pin4, J4-Pin6]`

Phys	BCM	Dir	Assignment	Notes / connector
14	—	GND	Ground	Additional even-side ground.
16	GPIO23	—	Spare / JP2 pin 3	Routed to JP2 pin 3. Not yet dedicated to a function. Available for button, indicator, or interface signal. `[JP2-Pin3, J3-Pin3]`
18	GPIO24	O	74HC595 RCLK (latch)	Output. Drives 74HC595 pin 12 (RCLK / storage register clock). A rising edge on this pin latches the shift register contents to the 74HC595 output pins (QA–QH), updating all 8 MOSFET gates simultaneously. Pulse this pin *after* each SPI1 byte transmission to update outputs. 33 Ω series resistor on PCB.
20	—	GND	Ground	Additional even-side ground.
22	GPIO25	O	MOSFET channel 10 (direct GPIO)	Drives gate of AO3400A MOSFET for LED channel 10. 500 Ω gate resistor + 10 kΩ bleed resistor to GND. See Section 6.1 for switching caution.
24	GPIO8	O	TFT CS → ILI9341 (SPI0 CE0)	**FIXED** by SoC (SPI0 CE0). Hardware-managed chip select. Asserted low automatically by SPI0 hardware during display transactions. 33 Ω series resistor on PCB. `[J1 ICSP area]`
26	GPIO7	O	MOSFET channel 9 (direct GPIO)	Drives gate of AO3400A MOSFET for LED channel 9. 500 Ω gate resistor + 10 kΩ bleed resistor to GND. Note: GPIO7 is SPI0 CE1 — not a conflict since SPI0 CE1 is unused. See Section 6.1 for switching caution.
28	GPIO1	—	PIN REMOVED — DO NOT USE	HAT ID EEPROM pin. Physical pin removed on this prototype.
30	—	GND	Ground	Additional even-side ground.
32	GPIO12	PWM	Buzzer (via MMBT3904 BJT)	**Hardware PWM channel 0-A (PWM0).** PWM signal drives base of MMBT3904 NPN BJT. BJT collector connects to buzzer (−) terminal. Buzzer (+) connects to 5 V rail. Flyback diode fitted: cathode (line-marked end) to buzzer(+)/5 V, anode to buzzer(−)/collector. 1 kΩ base resistor. See Section 5.5 for shared clock constraints with GPIO13.
34	—	GND	Ground	Additional even-side ground.

Phys	BCM	Dir	Assignment	Notes / connector
36	GPIO16	IO	1-Wire data bus	**Non-default 1-Wire pin.** Requires `dtoverlay=w1-gpio,gpiopin=16` in `/boot/firmware/config.txt` (BCM number, not physical pin). 4.7 kΩ pull-up to 3.3 V on PCB (hard requirement of 1-Wire protocol; internal RPi pull-up is too weak). Connects to J2-Pin2 (1-Wire probe plug) and J5–J8 Pin 8 (LED plugs). 100 nF ceramic decoupling at each device end of J5–J8 (fitted). See Section 5.4 for J2 cable note. `[J2-Pin2, J5-Pin8, J6-Pin8, J7-Pin8, J8-Pin8]`
38	GPIO20	O	SPI1 MOSI → 74HC595 SER	**FIXED** for SPI1 function. Drives 74HC595 pin 14 (SER / serial data input). Also exposed at J4-Pin5 for potential future SPI1 device. 33 Ω series resistor on PCB. `[74HC595 pin 14, J4-Pin5]`
39	—	GND	Ground	Odd-side ground (between pins 38 and 40).
40	GPIO21	O	SPI1 SCLK → 74HC595 SRCLK	**FIXED** for SPI1 function. Drives 74HC595 pin 11 (SRCLK / shift register clock). Each rising edge shifts one bit from SER into the shift register. Also exposed at J4-Pin3 for potential future SPI1 device. 33 Ω series resistor on PCB. `[74HC595 pin 11, J4-Pin3]`

74HC595 Shift Register — As-Built Pin Table

The 74HC595 provides 8 parallel output lines (QA–QH) from a 3-wire SPI1 serial interface. It controls LED channels 1–8 via AO3400A MOSFETs.

595 Pin	Name	Connects to	Function	Notes
1	QB	MOSFET ch 2	Parallel output B	500 kΩ to MOSFET gate, 10 kΩ bleed to GND. LED channel 2.
2	QC	MOSFET ch 3	Parallel output C	LED channel 3.
3	QD	MOSFET ch 4	Parallel output D	LED channel 4.
4	QE	MOSFET ch 5	Parallel output E	LED channel 5.
5	QF	MOSFET ch 6	Parallel output F	LED channel 6.
6	QG	MOSFET ch 7	Parallel output G	LED channel 7.
7	QH	MOSFET ch 8	Parallel output H	LED channel 8.

595 Pin	Name	Connects to	Function	Notes
8	GND	GND	Ground	**Caution:** 100 nF decoupling cap fitted at VCC (pin 16), but ground path to pin 8 is circuitous. Solder a short jumper wire directly from cap ground pad to pin 8 before testing to improve decoupling effectiveness.
9	QH'	Not connected	Serial cascade output	Not used. No daisy-chain in this design.
10	!SRCLR	3.3 V	Shift register clear	Tied high (3.3 V). Active-low — holding high keeps shift register enabled. To clear all outputs: pulse low in software (requires a GPIO if needed later).
11	SRCLK	RPi pin 40 / GPIO21	Shift register clock	SPI1 SCLK. Rising edge shifts one bit from SER into the register. 8 rising edges load a complete byte.
12	RCLK	RPi pin 18 / GPIO24	Storage register clock (latch)	Manual GPIO output (not a hardware SPI signal). Rising edge copies all 8 shift register bits to output pins simultaneously. Pulse *after* each SPI1 byte write. This separation prevents output glitching during bit-shifting.
13	!OE	GND	Output enable	Tied to GND (active-low). Outputs are permanently enabled. To tri-state outputs (e.g. power-down): a GPIO and a trace cut would be needed.
14	SER	RPi pin 38 / GPIO20	Serial data input	SPI1 MOSI. Receives one bit per SRCLK rising edge. MSB should be sent first for QA=bit7, QH=bit0 (verify with software).
15	QA	MOSFET ch 1	Parallel output A	First bit clocked in appears here after latch. LED channel 1. 500 kΩ to MOSFET gate, 10 kΩ bleed to GND.
16	VCC	3.3 V	Supply voltage	3.3 V from RPi rail. Output high level ≈3.0–3.3 V, sufficient for AO3400A gate threshold (1.5 V typical). 100 nF ceramic decoupling fitted (see pin 8 note above).

Connector Pin Tables

J1 — 16-pin header, Adafruit 1651 shield interface

Point-to-point wiring between J1 and the 1651 shield header pins. Series resistors (33 Ω) are placed on the PCB between the RPi header and J1 on all signal lines.

J1 Pin	RPi Pin/BCM	1651 Shield Pin	Signal / Notes
1	Pin 2 / 5 V	Arduino 5V / VIN	5 V supply to shield. 100 nF + 10 μF decoupling fitted.
2	Pin 3 / GPIO2	SDA	I2C SDA → TSC2007.
3	GND	GND	Ground.
4	Pin 5 / GPIO3	SCL	I2C SCL → TSC2007.
5	GND	GND	Ground.
6	—	—	Not connected.
7	Pin 31 / GPIO6	Arduino D9	TFT DC (Data/Command select).
8	Pin 29 / GPIO5	Arduino D4	SD Card CS (unassigned; spare).
9	Pin 33 / GPIO13	Arduino D3 / LITE	Backlight PWM. Solder LITE/#3 jumper on shield.
10	Pin 7 / GPIO4	Arduino D2 / TSIRQ	Touch interrupt (active-low). Solder TSIRQ/#2 jumper on shield.
11	Pin 23 / GPIO11	ICSP SCK	SPI0 SCLK.
12	Pin 21 / GPIO9	ICSP MISO	SPI0 MISO ← ILI9341.
13	Pin 19 / GPIO10	ICSP MOSI	SPI0 MOSI → ILI9341.
14	Pin 11 / GPIO17	—	Spare GPIO.
15	GND	GND	Ground.
16	Pin 37 / GPIO26	—	Spare GPIO.

1651 VddIO jumper — must be set before power-on: The shield's VddIO solder jumper selects the logic voltage reference for the onboard 74LVX245 level shifters. **Set to 3V** (solder left pad to centre pad). The RPi Zero 2 W is a 3.3 V logic device; incorrect setting risks damaged GPIO pins.

Shield jumpers to solder: LITE/#3 (enables backlight PWM control) and TSIRQ/#2 (enables touch interrupt).

J2 — 3-pin polarised, 1-Wire temperature probe

J2 Pin	Signal	Notes
1	3.3 V	Supply to 1-Wire device.
2	Data	1-Wire data. RPi pin 36 / GPIO16. 4.7 kΩ pull-up to 3.3 V on PCB.
3	GND	Ground.

J2 cable note: J2 connects to a 1-Wire temperature probe via a ≈1 metre cable. There is no decoupling capacitor at J2 on the PCB. The nearest decoupling caps on the 3.3 V rail are ≈5 inches away via J5–J8. The 4.7 kΩ pull-up resistor is on the PCB side. This arrangement is expected to work at standard 1-Wire speed (15.4 kbps), as the pull-up source impedance is the dominant factor, not proximity of bypass caps. If communication errors occur during testing, fit a 100 nF ceramic cap between pins 1 and 3 of J2 (3.3 V to GND) as the first corrective step.

J3 — 4-pin polarised, interface control (partial)

J3 Pin	Signal	Notes
1	GND	Ground.
2	3.3 V	3.3 V supply.
3	GPIO23	RPi pin 16. Spare / unassigned. Probably button or switch.
4	GPIO18	RPi pin 12 via **JP2 (closed)**. See jumper note below.

J4 — 6-pin polarised, interface control / SPI1 breakout

J4 Pin	Signal	Notes
1	3.3 V	3.3 V supply.
2	GND	Ground.
3	GPIO21	RPi pin 40. SPI1 SCLK. Also drives 74HC595 SRCLK.
4	GPIO19	RPi pin 35. SPI1 MISO. Not currently connected to a device.
5	GPIO20	RPi pin 38. SPI1 MOSI. Also drives 74HC595 SER.
6	GPIO18	RPi pin 12 via **JP1 (open)**. Not currently active.

JP1 / JP2 — solder jumpers for GPIO18 routing: GPIO18 is connected to both J3-Pin4 (via JP2) and J4-Pin6 (via JP1). **Current state: JP2 closed, JP1 open.** GPIO18 is therefore active only at J3-Pin4. **Do not close both jumpers simultaneously** — this would short J3-Pin4 directly to J4-Pin6 through GPIO18. GPIO18 is also the hardware SPI1 CE0 pin; see Section 5.2.

J5, J6, J7, J8 — 8-pin ribbon headers, LED banks

J5 and J6 carry identical signals (LED channels 1–5). J7 and J8 carry identical signals (LED channels 6–10). Duplicate headers allow either plug to connect to the same LED bank.

Pin	Signal	Notes
1	5 V (RPi pin 4)	LED anode supply. 100 nF ceramic decoupling near each bank.
2–6	LED channels	MOSFET drain-switched. See channel assignment table below.
7	GND	Ground return for LED loads.
8	1-Wire data	GPIO16 / pin 36. 100 nF ceramic cap at device end (fitted).

LED channel assignment across headers:

Header pin	J5, J6 channel	J7, J8 channel	Drive source
2	Ch 1	Ch 6	74HC595 QA / QB
3	Ch 2	Ch 7	74HC595 QB / QC
4	Ch 3	Ch 8	74HC595 QC / QD
5	Ch 4	Ch 9	74HC595 QD / GPIO7 (direct)
6	Ch 5	Ch 10	74HC595 QE / GPIO25 (direct)

Channel assignment across J5–J8 is inferred from the as-built description (channels 1–5 on J5/J6, channels 6–10 on J7/J8, 8 channels via 595, 2 via direct GPIO). Verify pin 2–6 assignments against the PCB during testing.

OS Configuration — `/boot/firmware/config.txt`

All settings below are required in `/boot/firmware/config.txt` (Raspberry Pi OS Bookworm and later; older releases use `/boot/config.txt`). Add or confirm each line before first power-on test.

SPI0 — ILI9341 display

```
dtparam=spi=on
```

Enables SPI0 (GPIO8–11). Required for ILI9341 display and optional SD card. CE0 (GPIO8) is managed automatically by the spidev driver.

SPI1 — 74HC595 shift register

```
dtoverlay=spi1-1cs,cs0_pin=24
```

SPI1 CE0 conflict — read carefully before configuring.

The default `spi1-1cs` overlay assigns **GPIO18 as CE0**. GPIO18 in this design is used for JP2/J3 routing — not for SPI1 CE0.

The 74HC595 does not use a hardware chip select at all. RCLK (GPIO24) is controlled manually in software to latch outputs.

The recommended approach is:

1. Load the overlay with a remapped CS pin that is not in use: `dtoverlay=spi1-1cs,cs0_pin=24` redirects CE0 to GPIO24. However, GPIO24 is already used for RCLK. A cleaner alternative: use `cs0_pin=X` where X is a confirmed spare GPIO (e.g. GPIO23/pin 16 if it remains unassigned).
2. In Python, open SPI1 with `no_cs=True` via spidev, disabling hardware CE entirely:
```
import spidev
spi1 = spidev.SpiDev()
spi1.open(1, 0)
spi1.no_cs = True
spi1.max_speed_hz = 1_000_000
```
3. After each `spi1.writebytes([byte])`, pulse GPIO24 (RCLK) manually:
```
import RPi.GPIO as GPIO
GPIO.setmode(GPIO.BCM)
GPIO.setup(24, GPIO.OUT)

def shift_out(byte_val):
    spi1.writebytes([byte_val])
    GPIO.output(24, GPIO.HIGH)   # latch rising edge
    GPIO.output(24, GPIO.LOW)    # return low
```

This keeps GPIO18 free and gives precise control of the latch timing.

I2C — TSC2007 touch controller

```
dtparam=i2c_arm=on
```

Enables I2C bus 1 (GPIO2, GPIO3). TSC2007 default I2C address: `0x48`. Verify with `sudo i2cdetect -y 1` after wiring.

1-Wire — temperature probe on GPIO16

```
dtoverlay=w1-gpio,gpiopin=16
```

Non-default pin. The standard 1-Wire pin is GPIO4. This design uses **GPIO16 / physical pin 36**. The overlay parameter `gpiopin=` takes the **BCM number (16)**, not the physical pin number. Using `gpiopin=7` (the BCM number of physical pin 26, SPI0 CE1) would configure the wrong pin and leave GPIO16 inactive.

The 4.7 kΩ pull-up to 3.3 V on the PCB is a hard requirement. The RPi internal pull-up (≈50 kΩ) is insufficient for 1-Wire timing.

After configuration, probe devices will appear under: `/sys/bus/w1/devices/` Temperature sensors (DS18B20 etc.) register as `28-xxxxxxxxxxxx`. Read temperature: `cat /sys/bus/w1/devices/28-*/w1_slave`

Hardware PWM — GPIO12 (buzzer) and GPIO13 (backlight)

Hardware PWM requires no `config.txt` entry — it is always available on GPIO12, GPIO13, GPIO18, GPIO19. However, using it from Python requires either `pigpio` or direct register access. **Use `pigpio` — it handles all clock arithmetic.**

```
# Start the pigpio daemon (add to /etc/rc.local or use systemd):
sudo pigpiod

# Python usage:
import pigpio
pi = pigpio.pi()

# Set backlight first, then buzzer
# (pigpio optimises the shared clock for the last call;
#  setting buzzer last gives it priority for exact frequency)
pi.hardware_PWM(13, 2000, 700000)   # backlight: 2000 Hz, 70% duty
```

```
pi.hardware_PWM(12, 2048, 500000)   # buzzer: 2048 Hz, 50% duty

# Adjust buzzer frequency in real time (fine-grained, 1 Hz steps):
for freq in range(1500, 2500, 10):
    pi.hardware_PWM(12, freq, 500000)
    time.sleep(0.05)

# Silence buzzer without disturbing backlight:
pi.hardware_PWM(12, 2048, 0)        # duty=0 silences output

# Adjust backlight brightness (0-100%):
def set_backlight(percent):
    pi.hardware_PWM(13, 2000, percent * 10000)
```

Shared PWM clock constraint — GPIO12 and GPIO13.

GPIO12 and GPIO13 share the same hardware PWM clock divider (PWM0/PWM1 on the RP3A0). Their output frequencies are derived from a single base clock frequency. This means:

- You can set *independent duty cycles* on GPIO12 and GPIO13.
- You *cannot* set *completely independent frequencies* simultaneously — both frequencies must be achievable from the same base clock.
- `pigpio` handles this automatically: it finds the best shared clock for both requested frequencies and minimises error. Typical frequency accuracy is within 1% across the audio range.
- **Always set the backlight first, then the buzzer.** `pigpio` optimises the clock for the last call, giving the buzzer (the frequency-critical channel) priority.
- At 2000 Hz (backlight) and 2048 Hz (buzzer), `pigpio` will find a common base clock that satisfies both to within a few Hz. The backlight frequency may deviate slightly from 2000 Hz — this is imperceptible and does not affect LED brightness control.

Practical buzzer frequency range for CEM-1203(42):

- Rated resonant frequency: 2048 Hz (loudest output)
- Useful audible range: 800–4000 Hz
- Best efficiency / loudest: 1500–2500 Hz
- `pigpio` supports any integer Hz value; 1 Hz steps throughout the range

Complete `config.txt` additions summary

Add the following lines to `/boot/firmware/config.txt`:

```
# Colorimeter prototype configuration
dtparam=spi=on                          # SPI0: ILI9341 display
dtparam=i2c_arm=on                      # I2C1: TSC2007 touch
dtoverlay=w1-gpio,gpiopin=16            # 1-Wire: GPIO16 (non-default)
dtoverlay=spi1-1cs,cs0_pin=X           # SPI1: 74HC595 (replace X with spare GPIO)
```

Enable and start `pigpio` daemon for hardware PWM:

```
sudo systemctl enable pigpiod
sudo systemctl start pigpiod
```

Hardware Cautions for Testing

74HC595 gate resistors — slow MOSFET switching

The 74HC595 output pins connect to AO3400A MOSFET gates through **500 kΩ** resistors (10 kΩ bleed resistors to GND). The direct-GPIO MOSFETs (channels 9 and 10) use 500 Ω gate resistors.

Effect of 500 kΩ gate resistor: The AO3400A input capacitance (C_{iss}) is approximately 130 pF.

Time constant $\tau = 500\,\text{k}\Omega \times 130\,\text{pF} = 65\,\mu\text{s}$. The MOSFET reaches full enhancement (Vgs ≈ 3.3 V) after approximately $5\tau = 325\,\mu\text{s}$. Turn-off time through the 10 kΩ bleed is approximately $5 \times 10\,\text{k}\Omega \times 130\,\text{pF} = 6.5\,\mu\text{s}$ (faster).

Practical impact: For LED switching at 100–5000 ms intervals, the 325 µs turn-on delay is negligible (<0.3% of the shortest cycle). The MOSFET will briefly pass through its linear region during turn-on, dissipating some heat. At your duty cycles (<10 cycles per analysis, <1 analysis per 10 minutes), this is not a thermal concern.

Observation to watch for during testing: LED onset may appear slightly soft (not an instantaneous step) on an oscilloscope. If any MOSFETs feel warm to the touch after repeated cycling, note it for revision. The AO3400A is rated 5.7 A continuous drain current — it is not at risk of damage.

74HC595 decoupling ground path

The 100 nF decoupling capacitor at the 74HC595 VCC pin has a circuitous ground path back to pin 8 (GND). **Before testing:** solder a short jumper wire directly from the capacitor's ground pad to pin 8 of the 74HC595 to provide a low-inductance local ground return. Without this, the decoupling is partially ineffective at SPI clock frequencies.

SPI1 CE0 and GPIO18

See Section 5.2. The `spi1-1cs` overlay claims GPIO18 as CE0 by default. If the overlay is loaded without the `cs0_pin` parameter, GPIO18 will be driven by the kernel SPI driver, which will conflict with the JP2 routing to J3-Pin4. Configure the overlay correctly *before* first power-on with SPI1 active.

Buzzer flyback diode orientation

The flyback diode is fitted with the cathode (line-marked end) at buzzer(+) / 5 V, and the anode at buzzer(−) / MMBT3904 collector. This is the correct orientation for inductive kickback suppression: when the BJT switches off, the collapsing magnetic field in the buzzer coil drives the collector positive, forward-biasing the diode and clamping the voltage spike. **Verify diode orientation before first power-on.**

Adafruit 1651 Shield Pin Mapping

1651 Pin (Arduino label)	RPi Pin / BCM	Description	Notes
5V / VIN	Pin 2 / 5 V	Board power in	5 V from RPi. Onboard regulator supplies 3.3 V to ILI9341 and TSC2007.
GND	GND	Ground	J1-Pin3, Pin5, Pin15.
ICSP MOSI	Pin 19 / GPIO10	SPI0 MOSI → ILI9341	**FIXED**. J1-Pin13. 33 Ω series resistor.
ICSP MISO	Pin 21 / GPIO9	SPI0 MISO ← ILI9341	**FIXED**. J1-Pin12. 33 Ω series resistor.
ICSP SCK	Pin 23 / GPIO11	SPI0 clock	**FIXED**. J1-Pin11. 33 Ω series resistor.
D10 — TFT CS	Pin 24 / GPIO8	Display chip select (SPI0 CE0)	**FIXED** (SPI0 CE0). Hardware-managed. 33 Ω series resistor.

1651 Pin (Arduino label)	RPi Pin / BCM	Description	Notes
D9 — TFT DC	Pin 31 / GPIO6	Data/Command select	Soft. J1-Pin7. 33 Ω series resistor.
D4 — SD CS	Pin 29 / GPIO5	microSD chip select	Unassigned. J1-Pin8. Wire present; not yet driven in software.
SDA	Pin 3 / GPIO2	I2C SDA → TSC2007	**FIXED**. Shield has onboard 4.7 kΩ pull-up. J1-Pin2.
SCL	Pin 5 / GPIO3	I2C SCL → TSC2007	**FIXED**. Shield has onboard 4.7 kΩ pull-up. J1-Pin4.
D2 — TSIRQ	Pin 7 / GPIO4	Touch interrupt (active-low)	Soft. Solder TSIRQ/#2 jumper on shield. RPi internal pull-up in software. J1-Pin10.
D3 — LITE	Pin 33 / GPIO13	Backlight PWM	Soft. Solder LITE/#3 jumper on shield. Hardware PWM1. J1-Pin9.
VddIO jumper	**CRITICAL:** Set to **3V** (solder left pad to centre pad). Selects 3.3 V as logic reference for onboard level shifters. Incorrect setting (5V) risks permanent GPIO damage.		

Spare and Reserved Pins

Physical	BCM	Status
Pin 11	GPIO17	Spare. Routed to J1-Pin14. No function assigned.
Pin 13	GPIO27	No connect. Available.
Pin 15	GPIO22	No connect. Available.
Pin 16	GPIO23	Routed to JP2-Pin3 and J3-Pin3. Function TBD (button/switch likely).
Pin 18	GPIO24	74HC595 RCLK (assigned).
Pin 29	GPIO5	Spare. Routed to J1-Pin8 (SD CS). Not yet driven in software.
Pin 35	GPIO19	SPI1 MISO breakout at J4-Pin4. No device connected.
Pin 37	GPIO26	Spare. Routed to J1-Pin16. No function assigned.
Pin 27	GPIO0	EEPROM. Physical pin removed. DO NOT USE.
Pin 28	GPIO1	EEPROM. Physical pin removed. DO NOT USE.
Pin 8	GPIO14	UART TX. Leave unconnected.
Pin 10	GPIO15	UART RX. Leave unconnected.